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Chapter 1: Governing Equations in Mantle Dynamics

1.1 Mechanics of Mantle Dynamics

Fluid dynamics of the mixture of two phase flow has been extensively discussed. We will disscuss the three governing equations of mass and momentum referring to the similar discussion in Drew and Segel (1971), Mckenzie (1984) and Katz (2022). Considering a conservation equation for given volumetric variable γ in a representative volume element V,

$$\frac{d}{dt}\int_V \gamma d\mathbf{v} = -\int_{\partial V} \mathbf{f}\cdot d\sigma + \int_V S d\mathbf{v}, \tag{1.1}$$

where $\mathbf{f}$ represents flux travelling across the boundary of a representative volume element V. The minus sign indicates that the positive direction aligns with normal unit vector pointing outwards. S, source/sink term, represents the volumetric production rate of γ. We can apply Gauss's theorem to obtain differential form,

$$\frac{\partial \gamma}{\partial t} + \nabla\cdot\mathbf{f} = S. \tag{1.2}$$

We also define the quantity of phase fraction rigorously with the help of an indicator function admitting that multiple phase cannot coexist at the same spatial point, which are regarded as immiscible phases. We define indicator function,

$$i(\mathbf{x}) = \begin{cases} 1 & \text{phase}_i, \\ 0 & \text{not phase}_i. \end{cases} \tag{1.3}$$

We then define macroscopic phase fraction function with the help of (1.3),

$$\phi_i = \frac{1}{\delta\mathbf{v}}\int_{\delta\mathbf{v}} i(\mathbf{x}) d\mathbf{v}. \tag{1.4}$$

In the beginning of my discussion, we will discuss two immiscible phases liquid magma and solid mantle. Phase fraction of liquid magam is labelled as ϕ_l and phase fraction of solid mantle is labelled as ϕ_s, where $\phi_l + \phi_s = 1$ is always attained. Phase fraction can also be regarded as volume fraction in macroscopic sense. Substituting bulk density $\overline{\rho} = \phi_l \rho_l + \phi_s \rho_s$ for generalized volumetric variable γ in equation (1.1), we can obtain conservation law for bulk mass,

$$\frac{\partial \overline{\rho}}{\partial t} + \nabla \cdot \overline{\rho \mathbf{v}} = 0, \tag{1.5}$$

with the help of Boussinesq approximation ignoring compressibility of each individual phase, and extend Boussinesq approximation further applying to density difference between each phase except body force term. We can write continuous function for two phase mixture simply as,

$$\nabla \cdot \overline{\mathbf{v}} = 0. \tag{1.6}$$

Substituing phase averaged momentum $\overline{\rho \mathbf{v}}$, we then write conservation of momentum with phase averaged term,

$$\frac{\partial \overline{\rho \mathbf{v}}}{\partial t} + \nabla \cdot \overline{\rho \mathbf{v} \otimes \mathbf{v}} = \overline{\rho} \mathbf{g} + \nabla \cdot \overline{\tau} - \nabla \overline{p}, \tag{1.7}$$

where τ represents deviatoric stress. Ignoring change of inertia term due to near zero Reynolds number in magma and mantle,

$$\overline{\rho} \mathbf{g} + \nabla \cdot \overline{\tau} - \nabla \overline{p} = 0, \tag{1.8}$$

which represents the bulk conservation of momentum. This buld equation cannot determine two different velocities at the same time, hence, we need additional relationship governing the behavior of liquid through a porous media. We now take a closer look at the interphase force between solid and liquid phases. We will provide a brief overview of how the individual equations governing the liquid and solid phases are derived, while more detailed and rigorous proofs can be found in the mixture theory literature like Bowen (1980). We can write momentum balances for each individual phase as:

$$\begin{aligned} 0 &= \phi_l \rho_l \mathbf{g} + \nabla \cdot (\phi_l \tau_l) - \nabla(\phi_l p_l) + \mathbf{I}_l, \\ 0 &= \phi_s \rho_s \mathbf{g} + \nabla \cdot (\phi_s \tau_s) - \nabla(\phi_s p_s) - \mathbf{I}_s, \end{aligned} \tag{1.9}$$

where $\mathbf{I}_l + \mathbf{I}_s = 0$ is the interphase force between solid and liquid phases. We ignore deviatoric stress in liquid and keep isotropic stress only. We balance the remaining stress with viscous drag force and equilibrium term. We can reach the following equation aligning with Darcy's Law:

$$\phi_l(\mathbf{v}_l - \mathbf{v}_s) + \frac{k_{\phi_l}}{\mu_l}(\nabla p_l - \rho_f \mathbf{g}) = 0. \tag{1.10}$$

1.1.1 Compaction

We have an additional yet unique relationship regulating melt migration in mantle dynamics. We now revisit the continuous equation (1.6), with the two phases considered separately taking melting into consideration

$$\begin{aligned} \frac{\partial \phi_l}{\partial t} + \nabla \cdot (\phi_l \mathbf{v}_l) = \Gamma, \\ \frac{\partial (1-\phi_l)}{\partial t} + \nabla \cdot ((1-\phi_l)\mathbf{v}_s) = -\Gamma, \end{aligned} \tag{1.11}$$

where Γ is the unknown melting rate. We now focus on the solid phase equation

$$\frac{\partial \phi_l}{\partial t} + \mathbf{v}_s \cdot \nabla \phi_l = \Gamma + (1-\phi_l)\nabla \cdot \mathbf{v}_s, \tag{1.12}$$

where the divergence term $\nabla \cdot \mathbf{v}_s$ is often referred to as the compaction rate.

$$\mu_s \nabla \cdot \mathbf{v}_s - \phi_l(p_l - p_s) = 0, \tag{1.13}$$

which discribes compaction phenomenon.

1.1.2 Full System

Combining equations (1.6), (1.8), (1.10) and (1.17), We finally arive with the full flow mechanical system,

$$\phi_l(\mathbf{v}_l - \mathbf{v}_s) + \frac{k_{\phi_l}}{\mu_l}(\nabla p_l - \rho_f \mathbf{g}) = 0, \tag{1.14}$$

$$\nabla \cdot (\phi_l \mathbf{v}_l + \phi_s \mathbf{v}_s) = 0, \tag{1.15}$$

$$\nabla \bar{p} - \nabla \cdot \tau(\mathbf{v}_s) + \bar{\rho}\mathbf{g} = 0, \tag{1.16}$$

$$\mu_s \nabla \cdot \mathbf{v}_s - \phi_l(p_l - p_s) = 0, \tag{1.17}$$

where the deviatoric stress of the solid is

$$\tau(\mathbf{v}_s) = 2\mu_s\phi_s\big(\mathcal{D}\mathbf{v}_s - \tfrac{1}{3}\nabla \cdot \mathbf{v}_s\mathbf{I}\big), \tag{1.18}$$

and $\mathcal{D}\mathbf{v}_s = \frac{1}{2}(\nabla\mathbf{v}_s + \nabla\mathbf{v}_s^T)$ is the symmetric gradient. Moreover, the relative permeability is $k_{\phi_l} = k_0\phi_l^{2+2\Theta}$, where $\Theta \in [0, 1/2]$. μ_s and μ_l are dynamic viscosities for solid and liquid correspondingly.

1.1.3 Scaled Formulation

The Darcy part will mathematically degenerate in regions where the porosity is zero. To handle this issue, we use the formulation developed previously Arbogast et al. (2017). We first define pressure potentials as,

$$q_l = p_l - \rho_l g z \quad \text{and} \quad q_s = p_s - \rho_l g z, \tag{1.19}$$

where $g = |\mathbf{g}|$ and z is the depth, so $\mathbf{g} = g\nabla z$. Note that ρ_f is used to define both potentials, so that with $q = \bar{q} = \bar{p} - \rho_l g z$,

$$p_l - p_s = q_l - q_s = \frac{1}{1 - \phi_l}(q_l - q).$$

We can then rewrite the system (1.14)–(1.17) omitting the subscript of liquid volumetric fraction,

$$
\begin{aligned}
\phi \mathbf{v}_r + \frac{k_0 \phi^{2+2\Theta}}{\mu_l} \nabla q_l &= 0, && (1.20)\\
\mu_s \nabla \cdot (\phi \mathbf{v}_r) + \frac{\phi}{1-\phi}(q_l - q) &= 0, && (1.21)\\
\nabla q - \nabla \cdot \tau(\mathbf{v}_s) &= -(1-\phi)\rho_r \mathbf{g}, && (1.22)\\
\mu_s \nabla \cdot \mathbf{v}_s - \frac{\phi}{1-\phi}(q_l - q) &= 0. && (1.23)
\end{aligned}
$$

where relative velocity $\mathbf{v}_r = \mathbf{v}_l - \mathbf{v}_s$ and density difference $\rho_r = \rho_s - \rho_l$. We refer to ϕ as porosity in the later text. Scaled variables are defined,

$$
\tilde{\mathbf{v}}_r = \phi^{-\Theta} \mathbf{v}_r \quad \text{and} \quad \tilde{q}_l = \phi^{1/2} q_l, \tag{1.24}
$$

and the problem is reformulated as

$$
\begin{aligned}
\tilde{\mathbf{v}}_r + \frac{k_0 \phi^{1+\Theta}}{\mu_f} \nabla (\phi^{-1/2} \tilde{q}_l) &= 0, && (1.25)\\
\mu_s \phi^{-1/2} \nabla \cdot (\phi^{1+\Theta} \tilde{\mathbf{v}}_r) + \frac{1}{1-\phi}(\tilde{q}_l - \phi^{1/2} q) &= 0, && (1.26)\\
\nabla q - \nabla \cdot \tau(\mathbf{v}_s) &= -(1-\phi)\rho_r \mathbf{g}, && (1.27)\\
\mu_s \nabla \cdot \mathbf{v}_s - \frac{\phi^{1/2}}{1-\phi}(\tilde{q}_l - \phi^{1/2} q) &= 0, && (1.28)
\end{aligned}
$$

where the new scaled relative velocity maintain numerical stability in zero prosority region.

1.2 Conservation of Chemical Species and Energy

Chemical compositions of mantle and magma is very complicate and very hard to trace at the same time. We simplify the model to a two-phase system with two species. We negelect detailed description of chemical reaction process and effect on the Gibbs free energies. Similar discussion can be found in extensive literatures like Spiegelman and Elliott (1993), Hewitt and Fowler (2008), Katz (2008) and Hesse et al. (2022). Both two species can be solid or liquid. Two species are immiscible as solid but miscible as liquid. We consider a homogenous density ρ_s for solid phase and ρ_l for liquid phase. With extened Boussinesq approximation later assuming no density difference except in body force term, we can make further assumption that $\rho_l = \rho_s = \rho$ Within the representative volume element V, $\gamma_{i,j}$ represents one quantity

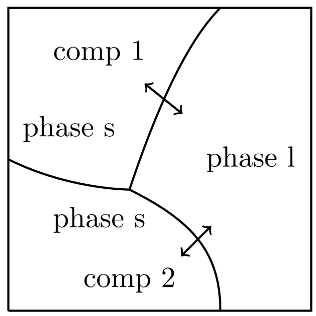

Figure 1.1: An representative volume element (RVE).

of the specie i presenting in phase j, where $i \in 1, 2$ and $j \in \{s, l\}$. The mass fraction $c_{i,j}$ of a given specie i in phase j is often regarded as concentration. Hence, we can write transport equations for a individual specie i assuming a pseudo melting term Γ,

$$\frac{\partial}{\partial t}\left(\phi_l \rho_l c_{i,l}\right) + \nabla \cdot \left(\phi_l \rho_l c_{i,l} \mathbf{v}_l - \phi_l \rho_l \mathcal{D}_{i,l} \nabla c_{i,l}\right) = \Gamma_i, \tag{1.29}$$

$$\frac{\partial}{\partial t}\left(\phi_s \rho_s c_{i,s}\right) + \nabla \cdot \left(\phi_s \rho_l c_{i,s} \mathbf{v}_s - \phi_s \rho_s \mathcal{D}_{i,s} \nabla c_{i,s}\right) = -\Gamma_i, \tag{1.30}$$

where we can safely get rid of dissusion term in solid phase. Summing up equation1.29 and equation1.30, we can write a advection-diffusion equation in phase averaged volumetric bulk variable,

$$\frac{\partial \overline{\rho c_i}}{\partial t} + \nabla \cdot \left(\overline{\rho c_i \mathbf{v}} - \phi_l \rho_l \mathcal{D}_{i,l} \nabla c_{i,l}\right) = 0, \tag{1.31}$$

where the phase averaged volumetric bulk variable are defined as $\overline{\rho c_i} = \sum_{j\in\{s,l\}} \phi_j \rho_j c_{i,j}$ and $\overline{\rho c_i \mathbf{v}} = \sum_{j\in\{s,l\}} \phi_j \rho_j c_{i,j} \mathbf{v}_j$. The subscript i will be omitted since only one chemical specie is traced in this two-component system. Extending Boussinesq approximation

$\rho_l = \rho_s = \overline{\rho}$, the specie transport equation can be simplified as,

$$\frac{\partial \overline{c}}{\partial t} + \nabla \cdot \left(\overline{c\mathbf{v}} - \phi_l \mathcal{D}_l \nabla c_l\right) = 0. \tag{1.32}$$

We write energy balance equation in the phase-averaged internal energy form first,

$$\frac{\partial \overline{\rho u}}{\partial t} + \nabla \cdot \overline{\rho u \mathbf{v}} - \nabla \cdot \overline{K} \nabla T = \overline{\rho \mathcal{Q}} + \nabla \cdot \overline{\sigma \cdot \mathbf{v}} + \overline{\rho \mathbf{v}} \cdot \mathbf{g}. \tag{1.33}$$

On the left hand side, we have time dependent rate of internal energy chage, advection term and diffusion term. On the right hand side, we have heating/cooling source $\mathcal{Q}$, work exerted by stress and work againt gravity g. In a thermochemical equilibrium scenerio, we are replacing internal energy with enthalpy. We can write energy balancing equation neglecting additional heating/cooling term $\mathcal{Q}$ and dissipative heating following the discussion in Mckenzie (1984),

$$\frac{\partial \overline{\rho h}}{\partial t} + \rho c_p \nabla \cdot \overline{\mathbf{v}} T = \rho L \nabla \cdot \phi_s \mathbf{v}_s + \overline{K} \nabla^2 T + \rho \alpha_\rho T \overline{\mathbf{v}} \cdot \mathbf{g}, \tag{1.34}$$

where specific heat capacity c_p is used for both components in solid and liquid phase. Similar but slighted altered form was also discussed in Katz (2008), where adiabatic cooling effect was added in terms of potential temperatureMckenzie and Bickle (1988). We now divide ρc_p on both sides of the equation considering Boussinessq approximation,

$$\frac{\partial H}{\partial t} + \nabla \cdot \overline{\mathbf{v}} T = \frac{L}{c_p} \nabla \cdot \phi_s \mathbf{v}_s + \kappa \nabla^2 T + \frac{\alpha_\rho}{c_p} T \overline{\mathbf{v}} \cdot \mathbf{g}, \tag{1.35}$$

where $H = \overline{h}/c_p$ is the bulk specific enthalpy and $\kappa = \overline{K}/\rho c_p$ is the heat diffusivity.

1.2.1 Effective Velocity

Effective velocity in equilibrium transport for concentration within liquid phase is discussed in Spiegelman (1996). We also introduce our effective velocity, but instead of transporting concentration in liquid(magma), we transport bulk concentration. Define partition coefficient $D_i = c_{i,s}/c_{i,f}$ and

$$(\phi_l + \phi_s D_i)c_{i,l} = \bar{c}_i, \tag{1.36}$$

$$(\phi_l/D_i + \phi_s)c_{i,s} = \bar{c}_i. \tag{1.37}$$

Substitute $c_{i,l}$ and $c_{i,s}$ with equation (1.36) and equation (1.37) into bulk velocity $\overline{c_i\mathbf{v}} = \phi_s\mathbf{v}_s c_{i,s} + \phi_l\mathbf{v}_l c_{i,l}$,

$$\overline{c_i\mathbf{v}} = \phi_s\mathbf{v}_s c_{i,s} + \phi_l\mathbf{v}_l c_{i,l} = \left(\frac{\phi_l\mathbf{v}_l + D\phi_s\mathbf{v}_s}{\phi_l + \phi_s D}\right)\bar{c}_i. \tag{1.38}$$

Hence, we get the proper expression for the effective velocity as $\mathbf{v}_{\text{eff}} = \frac{\phi_l\mathbf{v}_l + D_i\phi_s\mathbf{v}_s}{\phi_l + \phi_s D_i}$, where the partition coefficient D can be obtained through phase behavior package. If there is no melting $\phi_l = 0$, we can reterive $\mathbf{v}_{\text{eff}} = \mathbf{v}_s$. If there is all melting $\phi_l = 1$, we can obtain $\mathbf{v}_{\text{eff}} = \mathbf{v}_l$. Hence, the definition of effective velocity is well defined in those two boundary scenerios. We rewrite the transport of specie with effective velocity,

$$\frac{\partial C}{\partial t} = \nabla \cdot (\mathbf{v}_{\text{eff}} C - \phi_l \mathcal{D}_l \nabla c_l), = 0 \tag{1.39}$$

where $C = \bar{c}$ represents volumetric phase averaged (bulk) concentration of species.

1.3 Eutectic Phase Behavior

The enthalpy method requires an explicit relationship between temperature and enthalpy. Embracing the fact that mid-ocean ridge forms at the place where temperature is generally lower, eutectic melting, where melting temperature is signaficantly lowered with the presence of the secondary specie, is a suitable model for modelling melting behavior at certain location. The binary eutectic phase behavior describes the chemical behavior of two immiscible crystals from a completely miscible solution. On a typical eutectic phase behavior diagram, we can locate any phase situation using transported enthalpy and concentration. Binary eutectic melting has been applied to water-salt-brine system in glacier Hesse et al. (2022).

1.3.1 Eutectic Phase Parameters

We approximate the melting temperature and eutectic temperature at a given pressure using Calusius-Clapeyron relationship,

$$\begin{aligned} T_m &= T_{m0} + \nu^{-1} P, \\ T_e &= T_{e0} + \nu^{-1} P, \end{aligned} \tag{1.40}$$

where ν is the Clausius-Clapeyron constant.

In a eutectic system, we use bulk mass fraction and bulk enthalpy to locate the current state of a given system. $\mathcal{X}_i$ represents bulk mass fraction of specie i and $\mathcal{X}_{i,e}$ represents eutectic mass fraction of specie i in liquid. The subscript i will be omitted in the later context since only one specie is being traced in transport equation. Some assumptions are made for partially molten mantle system.

- Assuming thermodynamic equilibrium is achieved at every time step that phase state can be calculated with phase diagram.
- The composition of the second specie in the liquid formed by eutectic melting $\mathcal{X}_e$ is constant.
- Melting temperature T_m and eutectic temperature T_e are functions to pressure $T_m(P)$ and $T_e(P)$.
- Boussinesq approximation is extended to phase behavior as well $\rho_l = \rho_s$.
- Assuming no density difference between two solid components.
- Assuming constant thermal expansion and specific heat capacity.

We can now relate bulk mass fraction with bulk concentration directly,

$$\mathcal{X} = \frac{\phi_l \rho_l c_l + \phi_s \rho_s c_s}{\phi_l \rho_l + \phi_s \rho_s} = \phi_l c_l + \phi_s c_s = C = \phi_2 + \phi_l c_l. \tag{1.41}$$

We relate bulk specific enthalpy with temperature and latent heat again.

$$H = T + \phi_l L/c_p. \tag{1.42}$$

We further rescale by dividing melting temperature $\Delta T = T_{m0} - T_{e0}$ at standard atmospheric pressure forming a dimensionless relationship,

$$\check{H} = \check{T} + \phi_l \check{L}, \tag{1.43}$$

where $\check{L} = \frac{L}{c_p \Delta T}$ is the dimensionless latent heat.

1.3.2 Eutectic Phase Split

For simplicity, we are now assuming a linear eutectic phase split between different phase regions. We can draw the eutectic phase diagram separating different phase regions with straight lines at a fixed pressure. We now describe all six phase

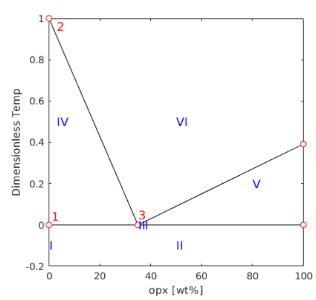

Figure 1.2: Linear Eutectic Phase Spliting.

regions depicting in the phase split diagram above,

I Single-phase sub-solidus : solid1

II Two-phase sub-solidus : solid1 + solid2

III Three-phase eutectic : solid1 + solid2 + melt

IV Two-phase super-eutectic : solid1 + melt

V Two-phase super-eutectic : solid2 + melt

VI Single-phase super-liquidus : melt

In the following discussion, we will restrict all the discussion to a fixed pressure.

Region I. Pure component 1. There is no eutectic phase behavior in this region. The conditions to be in region I are

$$C = 0 \quad \check{H} \leq \check{T}_m, \tag{1.44}$$

and we can determine phase fraction, mass composition and current temperature,

$$\phi_1 = 1, \quad \phi_2 = \phi_l = 0 \quad \text{and} \quad \check{T} = \check{H}. \tag{1.45}$$

Region II. Sub-solidus: two components coexist as solid phase. In this phase region, temperature hasn't reached eutectic melting temperature. The conditions to be in region II are

$$C \neq 0 \quad \check{H} \leq \check{T}_e. \tag{1.46}$$

We can determine phase fraction, mass composition and current temperature,

$$\phi_2 = C \quad \phi_1 = 1 - \phi_2 \quad \phi_l = 0 \quad \text{and} \quad \check{T} = \check{H}. \tag{1.47}$$

Region III. Two-phase eutectic: The system starts melting at eutectic temperature. Both solid component 1 and component 2 and melt coexist at the same time. Temperature fixs until component 2 exhausts leaving eutectic melting. Mass composition of component 2 in the melt is also fixed as $\mathcal{X}_e$. The conditions to be in region III are We can determine phase fraction, mass composition and current temperature,

$$\phi_l = \frac{\check{H} - \check{T}_e}{\check{L}} \quad \phi_2 = C - \phi_l \mathcal{X}_e \quad \phi_1 = 1 - \phi_l - \phi_2 \quad \text{and} \quad \check{T} = \check{T}_e. \tag{1.48}$$

Region IV. Super-eutectic two phase region. All component 2 in solid has been exhausted. The mass composition of component 2 in liquid c_f is now approximated with a relation between mass fraction and temperature. Subscript indicating component

2 has been dropped since only one component is being tracked. We can determine phase fraction first,

$$\phi_2 = 0 \quad \phi_l = \frac{C}{c_l(\check{T})}. \tag{1.49}$$

The corresponding temperature should be determined by solving the following equation.

$$F(\check{T}) = \check{H} - \check{T} - \frac{C}{c_l(\check{T})}\check{L} = 0. \tag{1.50}$$

We can solve this nonlinear function with Newton iteration. However, we are approximating with a linear relationship $c_l = \mathcal{X}_e(\check{T}_m - \check{T}))$ where $c_l > C$ is imposed for the fact that there is component 1 remaining as solid. Now we can solve a simple quadratic equation,

$$\check{T}^2 - (\check{H} + \check{T}_m)\check{T} + \check{T}_m\check{H} - CL/\mathcal{X}_e = 0. \tag{1.51}$$

Omitting the root that exceeding melting temperature,

$$\begin{aligned}\check{T} &= \frac{\check{H} + \check{T}_m - \sqrt{(\check{H} + \check{T}_m)^2 - 4(\check{T}_m\check{H} - CL/\mathcal{X}_e)}}{2} \\ &= \frac{\check{H} + \check{T}_m - \sqrt{(\check{H} - \check{T}_m)^2 + 4CL/\mathcal{X}_e}}{2}\end{aligned} \tag{1.52}$$

1.4 Nondimensionalization

Compaction and melting are two most important aspects in mantle dynamics. We nondimensionlize the scaled formulation with characteristic length scale related to compaction length and scaled characteristic Darcy speed of the melt,

$$l_0 = \big(\frac{k_0\mu_s}{\mu_l}\big)^{1/2} \quad \text{and} \quad u_0 = \frac{k_0\rho_r g}{\mu_l}. \tag{1.53}$$

We can derive corresponding characteristic time scale as $t_0 = l_0/u_0$, now we can write full set of dimensionless equations. We represent dimensionless variables with

Quantity	Value	Units
ρ	3000	$kg \cdot m^{-3}$
ρ_r	600	$kg \cdot m^{-3}$
α_ρ	3×10^{-5}	K^{-1}
c_p	1200	$J \cdot kg^{-1} \cdot K^{-1}$
L	5×10^5	$J \cdot kg^{-1}$
μ_l	1	Pascal $\cdot\, s$
μ_s	10^{19}	Pascal $\cdot\, s$
k_0	10^{-8}	m^2
T_{e0}	1480	K
T_{m0}	2000	K
ν	6.5×10^6	Pascal $\cdot\, K^{-1}$

Table 1.1: Flow Mechanics and Thermaldynamics Parameters

a ˇ mark,

$$\begin{aligned}
&\check{\mathbf{v}}_r + \phi^{1+\Theta}\nabla(\phi^{-1/2}\check{q}_l) &&= 0, &(1.54)\\
&\phi^{-1/2}\nabla\cdot(\phi^{1+\Theta}\check{\mathbf{v}}_r) + \frac{1}{1-\phi}(\check{q}_l - \phi^{1/2}\check{q}) &&= 0, &(1.55)\\
&\nabla\check{q} - \nabla\cdot\check{\tau}(\check{\mathbf{v}}_s) &&= -(1-\phi)\mathbf{n}, &(1.56)\\
&\nabla\cdot\check{\mathbf{v}}_s - \frac{\phi^{1/2}}{1-\phi}(\check{q}_l - \phi^{1/2}\check{q}) &&= 0, &(1.57)
\end{aligned}$$

where dimensionless deviatoric stress of the solid phase is $\check{\tau} = 2(1-\phi)(\mathcal{D}\check{\mathbf{v}}_s - 1/3\nabla\cdot\check{\mathbf{v}}_s\mathbf{I})$ and $\mathbf{n}$ is the unit direction vector pointing downwards. Now we align transport equations with dimensionless variables defined in mechanic system,

$$\begin{aligned}
\frac{\partial C}{\partial \check{t}} + \nabla\cdot(\check{\mathbf{v}}_{\text{eff}}C - \frac{\phi\mathcal{D}_l}{u_0 l_0}\nabla c_l) = 0,\\
\frac{\partial \check{H}}{\partial \check{t}} + \nabla\cdot\check{\bar{\mathbf{v}}}T - \check{L}\nabla\cdot\phi_s\check{\mathbf{v}}_s - \frac{\kappa}{u_0 l_0}\nabla^2\check{T} - \frac{\alpha_\rho l_0}{c_p}\check{T}\check{\bar{\mathbf{v}}}\cdot\mathbf{g} = 0.
\end{aligned} \quad (1.58)$$

Paramters used in the flow mechanics and thermaldynamic computations are listed in Table 1.1,

Chapter 2: Numerical Methods

I introduce a locally mass conserving numerical framework.

2.1 The Mixed Finite Element Method

2.1.1 Stokes Problem

We briefly describe solving static Stokes equation with mixed finite element scheme. We begin with strong form of the Stokes problem,

$$\begin{aligned} -\Delta \mathbf{u} + \nabla p &= \mathbf{f}, \\ \nabla \cdot \mathbf{u} &= 0. \end{aligned} \tag{2.1}$$

We derive weak form

$$\begin{aligned} (\nabla \mathbf{u}, \nabla \mathbf{v})_\Omega - (p, \nabla \cdot \mathbf{v})_\Omega &= (\mathbf{f}, \mathbf{v})_\Omega + \langle \tau^T \nabla \mathbf{u} \nu, \mathbf{v} \cdot \tau \rangle_{\Gamma_i}, \\ &\quad + \langle \nu^T \nabla \mathbf{u} \nu - p_0, \mathbf{v} \cdot \nu \rangle_{\Gamma_i} - \langle \nabla \mathbf{u}_0, \nabla \mathbf{v} \rangle_{\Gamma_u} \\ (\nabla \cdot \mathbf{u}, \omega) &= 0. \end{aligned} \tag{2.2}$$

Let stress $\mathbf{s} \cdot \tau = \tau^T \nabla \mathbf{u} \nu$ and $\mathbf{s} \cdot \nu = \nu^T \nabla \mathbf{u} \nu - p_0$, we have traction boundary condition term defined on Γ_i which is $< \mathbf{s}, \mathbf{v} >_{\Gamma_i}$. We employ a test space $\{\mathbf{v} \in H^1(\Omega) : \mathbf{v} = 0$ on $\Gamma_u\}$ here,

$$\begin{aligned} (\nabla \mathbf{u}_h, \nabla \mathbf{v}_h)_{\Omega_h} - (p_h, \nabla \cdot \mathbf{v}_h)_{\Omega_h} &= (\mathbf{f}, \mathbf{v}_h)_{\Omega_h} + \langle \mathbf{s}, \mathbf{v}_h \rangle_{\Gamma_{i,h}} - \langle \nabla \mathbf{u}_{0,h}, \nabla \mathbf{v}_h \rangle_{\Gamma_{u,h}}, \\ (\nabla \cdot \mathbf{u}, \omega) &= 0. \end{aligned} \tag{2.3}$$

We can separate normal and tangent part of stress on the boundary easily when physical boundary align with x-y axis. In the situation when a rectangular computational domain is used, we can mix stress and dirichlet boundary condition even further.

2.1.2 Darcy Problem

$$\begin{aligned} \mathbf{u} + \nabla p &= \mathbf{g} \\ \nabla \cdot \mathbf{u} &= 0 \end{aligned} \tag{2.4}$$

weak form

$$
\begin{aligned}
(\mathbf{u},\mathbf{v})_\Omega - (p,\nabla\cdot\mathbf{v})_\Omega = (\mathbf{g},\mathbf{v})_\Omega - \langle p_0, v_n\rangle_{\Gamma_i} - \langle \mathbf{u}_0,\mathbf{v}\rangle_{\Gamma_u}\\
(\nabla\cdot\mathbf{u},\omega) = 0
\end{aligned}
\tag{2.5}
$$

Test space $\{\mathbf{v}\in H(\mathrm{div},\Omega) : v_n = 0 \text{ on } \Gamma_u\}$.

2.1.3 Coupled Degenerate System

scaled nondimensionalized weak form let $\check{\sigma} = \mathcal{D}\mathbf{v} - \frac{1}{3}\nabla\cdot\mathbf{v}\mathrm{I}$

$$
\begin{aligned}
(2\phi_s\check{\sigma}(\mathbf{v}_{s,h}),\nabla\Psi_s) - (\check{q}_h,\nabla\cdot\Psi_s) &= -(\phi_s\check{\mathbf{f}},\Psi_s)\\
(\nabla\cdot\check{\mathbf{v}}_{s,h},\omega) + \Big(\frac{\hat{\phi}_f}{\phi_s}\check{q}_h,\omega\Big) - \Big(\frac{\hat{\phi}_f^{1/2}}{\phi_s}\check{q}_{f,h},\omega\Big) &= 0\\
(\check{\mathbf{v}}_{r,h},\Psi_r) - (\check{q}_{f,h},\hat{\phi}_f^{-1/2}\nabla\cdot(\phi_f^{1+\Theta}\check{\mathbf{v}}_r)) &= 0\\
(\hat{\phi}_f^{-1/2}\nabla\cdot(\phi_f^{1+\Theta}\check{\mathbf{v}}_r),\omega_f) + \Big(\frac{1}{\phi_s}\check{q}_{f,h},\omega_f\Big) - \Big(\frac{\hat{\phi}_f^{1/2}}{\phi_s}\check{q}_h,\omega_f\Big) &= 0
\end{aligned}
\tag{2.6}
$$

2.1.4 Degenerate Saddle Point Problem

The linear system corresponding to (insert eqcite) has the form

$$
\begin{bmatrix}
\mathbf{A}_s & -\mathbf{B}_s & \mathbf{0} & \mathbf{0}\\
\mathbf{B}_s^T & \mathbf{C}_s & \mathbf{0} & \mathbf{K}\\
\mathbf{0} & \mathbf{0} & \mathbf{A}_d & -\mathbf{B}_d\\
\mathbf{0} & \mathbf{K} & \mathbf{B}_d^T & \mathbf{C}_d
\end{bmatrix}
\begin{bmatrix}\mathbf{x}_s\\ \mathbf{y}_s\\ \mathbf{x}_d\\ \mathbf{y}_d\end{bmatrix}
=
\begin{bmatrix}\mathbf{f}_s\\ \mathbf{g}_s\\ \mathbf{f}_d\\ \mathbf{g}_d\end{bmatrix},
\tag{2.7}
$$

wherein the solution represents the degrees of freem of primary variables with respect to the bases of the finite element spaces. (Insert details of calculation of each blocked matrix). We form a double saddle point system by symmetrically permutate original linear system,

$$
\begin{bmatrix}
\mathbf{A}_s & \mathbf{0} & -\mathbf{B}_s & \mathbf{0}\\
\mathbf{0} & \mathbf{A}_d & \mathbf{0} & -\mathbf{B}_d\\
\mathbf{B}_s^T & \mathbf{0} & \mathbf{C}_s & \mathbf{K}\\
\mathbf{0} & \mathbf{B}_d^T & \mathbf{K} & \mathbf{C}_d
\end{bmatrix}
\begin{bmatrix}\mathbf{x}_s\\ \mathbf{x}_d\\ \mathbf{y}_s\\ \mathbf{y}_d\end{bmatrix}
=
\begin{bmatrix}\mathbf{f}_s\\ \mathbf{f}_d\\ \mathbf{g}_s\\ \mathbf{g}_d\end{bmatrix}.
\tag{2.8}
$$

We group these block matrix to form a concise version,

$$\mathbf{A} = \begin{bmatrix} \mathbf{A}_s & \mathbf{0} \\ \mathbf{0} & \mathbf{A}_d \end{bmatrix}, \quad \mathbf{B} = \begin{bmatrix} -\mathbf{B}_s & \mathbf{0} \\ \mathbf{0} & -\mathbf{B}_d \end{bmatrix}, \quad \mathbf{C} = \begin{bmatrix} -\mathbf{C}_s & -\mathbf{K} \\ -\mathbf{K} & -\mathbf{C}_d \end{bmatrix}, \quad \mathbf{F} = \begin{bmatrix} \mathbf{f}_s \\ \mathbf{f}_d \end{bmatrix}, \quad \mathbf{G} = \begin{bmatrix} -\mathbf{g}_s \\ -\mathbf{g}_d \end{bmatrix}, \tag{2.9}$$

$$\begin{bmatrix} \mathbf{A} & \mathbf{B} \\ \mathbf{B}^T & \mathbf{C} \end{bmatrix} \begin{bmatrix} \mathbf{x} \\ \mathbf{y} \end{bmatrix} = \begin{bmatrix} \mathbf{F} \\ \mathbf{G} \end{bmatrix}, \tag{2.10}$$

where $\mathbf{A}$ is a positive definite matrix. However, $\mathbf{B}$ is usually a singular matrix when no melting region presents. We will present a modified Uzawa iteration algorithm to deal with this coupled degenerate singular system. Inexact Uzawa algorithm has been introduced in the previous literature Elman and Golub (1994) and James H. Bramble (1997),

Algorithm 1 Preconditioned Inexact Uzawa Iteration

1: Initialize $\mathbf{x}_0 = \mathbf{0}$ and $\mathbf{y}_0 = \mathbf{0}$
2: Update $\mathbf{x}_{i+1} = \mathbf{x}_i + \mathbf{A}^{-1}(\mathbf{F} - (\mathbf{A}\mathbf{x}_i + \mathbf{B}\mathbf{y}_i))$
3: Obtain $\mathbf{r} = \mathbf{B}^T\mathbf{x}_{i+1} + \mathbf{C}\mathbf{y}_i - \mathbf{G}$
4: Obtain $\tilde{\mathbf{y}} = \operatorname{argmin}_{\mathbf{x}\in V} ||(\mathbf{B}^T\mathbf{A}^{-1}\mathbf{B} - \mathbf{C})^{-1}\tilde{\mathbf{y}} - \mathbf{r}||$
5: Update $\mathbf{y}_{i+1} = \mathbf{y}_i + \tilde{\mathbf{y}}$

2.2 The Multi-level Weighted Essentially Non-Oscillatory (ML-WENO) Method

We develop a accurate, versatile, robust and efficient finite volume weighted essentially non oscillatory (WENO) techniques for solving energy and composition conservation laws in Arbogast et al. (2024).

2.2.1 Stencil Polynomial

Given a stencil $\mathcal{S} = \{E_0, E_2, \ldots, E_k\}$ of the quasiuniform logically rectangular mesh. We can define an associated tensor product stencil polynomial $Q_{s,t} \in \mathbb{Q}_{(s,t)} := \mathbb{P}_{s-1}(x) \otimes \mathbb{P}_{t-1}(y)$, where s and t equal to the stencil sizes in the x- and y-directions, respectively. We scale our stencil polynomial individually as

$$Q_{s,t} = \sum_{p<s,q<t} c_{p,q} \Big(\frac{x - x_{\mathcal{S}}}{h_{\mathcal{S}}}\Big)^p \Big(\frac{y - y_{\mathcal{S}}}{h_{\mathcal{S}}}\Big)^q, \tag{2.11}$$

where $(x_{\mathcal{S}}, y_{\mathcal{S}})$ is the center point of the stencil $\Omega_{\mathcal{S}} = \cup_{E \in \mathcal{S}} \subset \mathbb{R}^2$, and $h_{\mathcal{S}} = \sqrt{\sum_k E_k/k}$ is the diameter of average cell size. Let $\xi = (x - x_{\mathcal{S}})/h_{\mathcal{S}}$ and $\eta = (y - y_{\mathcal{S}})/h_{\mathcal{S}}$ be the normalized varaibles, we can rewrite it into normalized version:

$$\hat{Q}_{s,t} = \sum_{p<s,q<t} c_{p,q} \xi^p \eta^q. \tag{2.12}$$

In finite volume scenerio, we are dealing with cell averaged value of the scalar transport variable

$$\overline{u}_E = \frac{1}{|E|} \int_E u(\mathbf{x}) d\mathbf{x} = \frac{1}{|E|} \int_E u(x, y) dx dy. \tag{2.13}$$

We can calculate coeffiicents $c_{p,q}$ by imposing a constraint that cell averaged value (2.13) is preserved through integral with stencil polynomial

$$\frac{1}{|E_k|} \int_{E_k} Q(\mathbf{x}) d\mathbf{x} = \overline{u}_{E_k}, \quad \forall E_k \in \mathcal{S}. \tag{2.14}$$

We will replace $\overline{u}_{E_k}$ with $\overline{u}_k$ in the following discussion for simplicity.

We can represent stencil polynomial with basis polynomials $Q_{s,t} = \sum_k \overline{u}_k \phi^k_{s,t}$, where basis polynomials can be calculated as

$$\frac{1}{|E_{k'}|} \int_{E_{k'}} \phi^k_{s,t}(\mathbf{x}) d\mathbf{x} = \frac{1}{|\hat{E}_{k'}|} \iint_{\hat{E}_{k'}} \hat{\phi}^k_{s,t}(\xi, \eta) d\xi d\eta = \begin{cases} 1 & k' = k \\ 0 & k' \neq k \end{cases} \quad \forall (E_{k'}, E_k) \in \mathcal{S}, \tag{2.15}$$

which would lead to a $(s \times t) \times (s \times t)$ non-singular linear system. We can now represent stencil polynomial with normalized basis polynomials as:

$$\hat{Q}_{s,t}(\xi, \eta) = \sum_k \overline{u}_k \hat{\phi}^k_{s,t}(\xi, \eta). \tag{2.16}$$

We define a linear projection operator $\pi_{\mathcal{S}}$ and let $\pi_{\mathcal{S}} u_{\mathcal{S}} = Q$ be a linear projection onto polynomial space W_p such that $\mathbb{Q}_{s,t} \subset W_p$. We estimate interpolation error with the Bramble-Hilbert Lemma Dupont and Scott (1980) and Bramble and Hilbert (1970) in $H_r(\Omega)$ space, where $r = s$ in x direction and $r = t$ in y-direction.

Lemma 2.1. *Let Ω_S be the domain covered by stencil $\mathcal{S}$ in $\mathbb{R}^2$ and let $u \in H^r(\Omega_{\mathcal{S}})$. We have:*

$$\|u - \pi_S u\|^2 \leq h^{2(r-1)} |u|^2_{H^r(\Omega_S)}. \tag{2.17}$$

Proof.

$$\|u - \pi_S u\| = \tag{2.18}$$

□

2.2.2 Stencil polynomial smoothness indicators

We define a smoothness indicator σ as a measurement of smoothness of $u(\mathbf{x})$ on the stencil S. The classic method from Jiang and Shu (1996) is defined by Sobolev seminorm as:

$$\sigma_P = \sum_{1 \leq |\alpha| \leq m} \int \int_{E_0} |E_0|^{|\alpha|-1} (\mathcal{D}^\alpha Q(x,y))^2 dxdy, \tag{2.19}$$

where $\alpha = (\alpha_1, \alpha_2)$ is a multi-index, $|\alpha| = \alpha_1 + \alpha_2$, and $\mathcal{D}$ is the derivative operator $\mathcal{D}^\alpha = \partial^{|\alpha|} / \partial x^{\alpha_1} \partial y^{\alpha_2}$. The Jiang-Shu smoothness indicator can be reformed with basis polynomials:

$$\begin{aligned}
\sigma_P &= \sum_{1 \leq |\alpha| \leq m} |E_0|^{|\alpha|-1} \iint_{E_0} (\mathcal{D}^\alpha Q(x,y))^2 dxdy \\
&= \sum_{1 \leq |\alpha| \leq m} \iint_{\hat{E}_0} (\mathcal{D}^\alpha \hat{Q}(\xi,\eta))^2 d\xi d\eta \\
&= \sum_{1 \leq |\alpha| \leq m} \iint_{\hat{E}_0} \big(\sum_{ij} \overline{u}_{ij} \mathcal{D}^\alpha \hat{\phi}_{i,j}(\xi,\eta) \big)^2 d\xi d\eta \\
&= \sum_{ij} \sum_{i'j'} \overline{u}_{ij} \overline{u}_{i'.j'} \sum_{1 \leq |\alpha| \leq m} \iint_{\hat{E}_{ij}} \mathcal{D}^\alpha \hat{\phi}_{ij}(\xi,\eta) \mathcal{D}^\alpha \hat{\phi}_{i'j'}(\xi,\eta) d\xi d\eta \\
&= \sum_{ij} \sum_{i'j'} \overline{u}_{ij} \overline{u}_{i'j'} \sigma_{ij}^{i'j'},
\end{aligned} \tag{2.20}$$

where $\sigma_{ij}^{i'j'}$ can be precomputed.

We proposed an alternative way of defining smoothness indicator in Huang et al. (2023).

$$\sigma_{\text{simp}} = \frac{1}{N_{\mathcal{S}} - 1} \sum_k (\overline{u}_k - \overline{u}_0)^2, \tag{2.21}$$

where $\overline{u}_0$ is the cell averaged solution in target cell. We define this simple smoothness indicator in order to reduce the compuational cost and to deal with distorted stencil on unstructured mesh. However, this simple smoothness indicator is not as robust as the classic Jiang-Shu smoothness indicator. Hence, we stick to the classic definition in the following discussion.

We observe that the smoothness indicator exhibit following convergence property that there exists some value $C > 0$ such that when $h \to 0^+$,

$$\sigma = \begin{cases} Ch^2 + \mathcal{O}(h^3) & \text{Smooth,} \\ \Theta(h) & \text{Not Smooth,} \end{cases} \tag{2.22}$$

where stencils are defined on quasi uniform mesh. This observation comes directly from the scaling property of Sobolev seminorm.

2.2.3 ML-WENO weighting

Given a target cell $E_{i,j}$, we can create a nonempty set of stencils $\mathcal{S}_l \ni E_{i,j}, l = 0, 1, \ldots, L$. The reconstruction of u on target cell Ei, j is a convex combination of stencil polynomials (2.16)

$$R(\mathbf{x}) = \sum_l \tilde{\omega}_l Q^l(\mathbf{x}), \quad \mathbf{x} \in E_{i,j}, \tag{2.23}$$

where $\sum_l \tilde{\omega}_l = 1$ is normalized nonlinear weights associated with stencils accordingly. We begin by arbitrarily chosing a set of linear weights $\{\omega_l\}$ and scale with stencil smoothness indicator as

$$\hat{\omega}_l = \frac{\omega_l}{(\sigma_l + \epsilon h_0^2)^{ar_l + n_l}}, \tag{2.24}$$

where $r_l = \max\{s, t\}$, $\epsilon = 1e - 4$ and $a > 0.5$ are picked as constants. The biasing factor $n_l = n(r_l)$ is used to biasing away further from constant level, whose smoothness

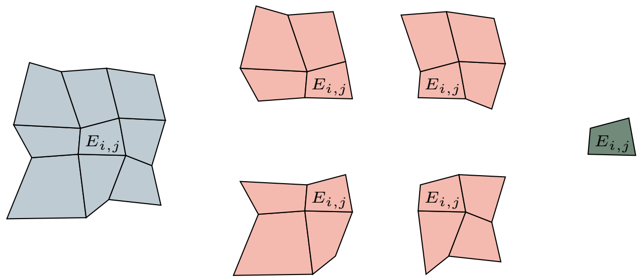

Figure 2.1: An example of (3,2,1) multi-level stencils combination on logically rectangular quadrilateral mesh.

indicator is always zero, when the solution is smooth. We pick

$$n(r_l) = n_l = \begin{cases} 1, & \text{if} \quad r_l = 1, \\ 3, & \text{if} \quad r_l = 2, \\ 4, & \text{if} \quad r_l \geq 3. \end{cases} \tag{2.25}$$

We then renormalize scaled weights

$$\tilde{\omega}_l = \frac{\hat{\omega}_l}{\sum_{l'} \hat{\omega}_{l'}}. \tag{2.26}$$

2.2.4 Boundary Treatment

2.2.5 Finite Volume Formulation with ML-WENO

2.2.6 Time Stepping

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